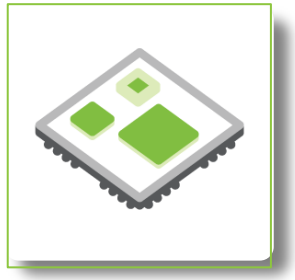
SERVER:
OPEN CHIPLET
ECONOMY



AI-native Virtual Chiplet Eco-systems: Industry best practices to accelerate Chiplet adoption

Thur Oct 15, 10:30am - 11:10am PT

Concourse Level - 211

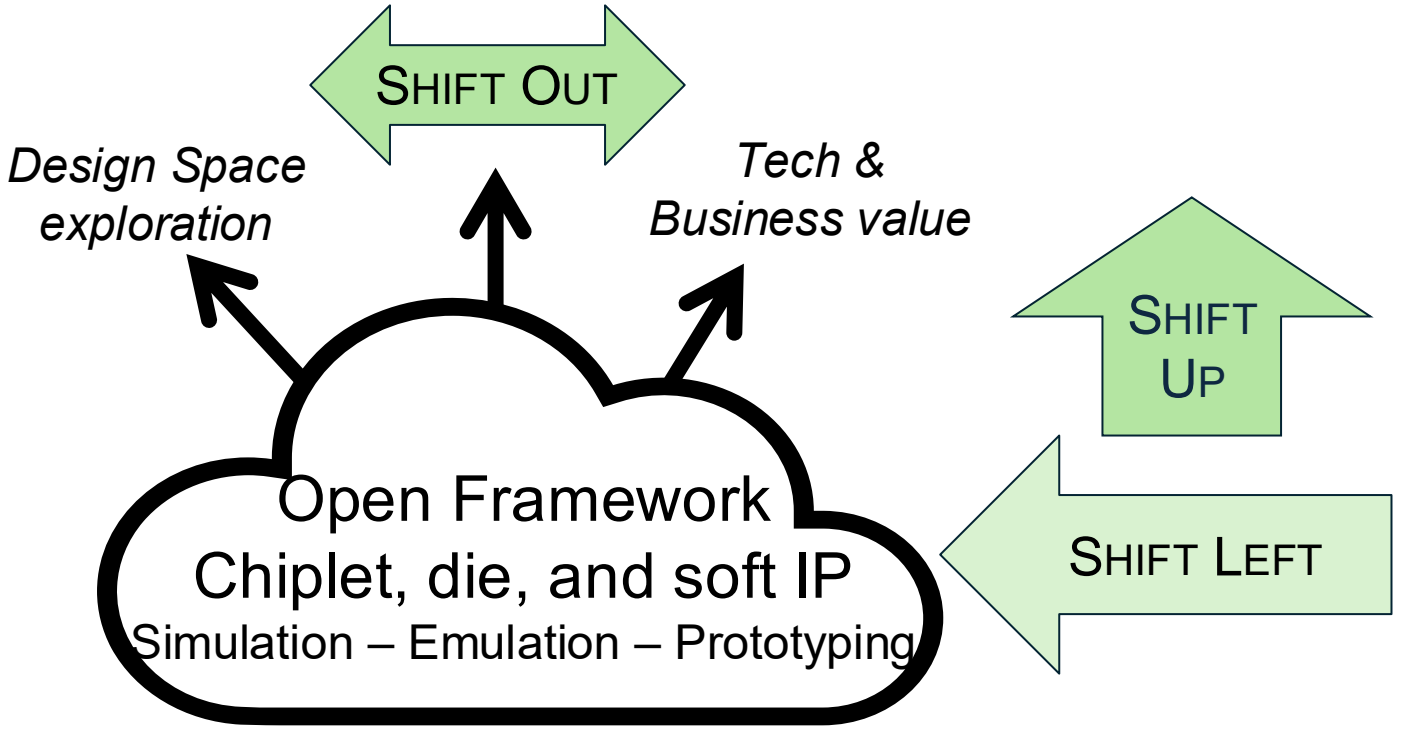
Raghu Shankar, Moderator

Entrepreneur

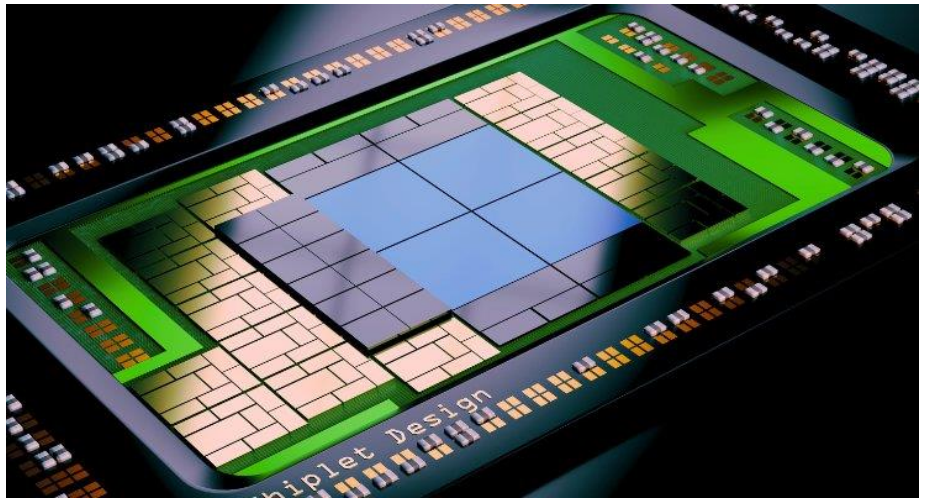
Expand the market, maximize business opportunity, and speed up time to solutions

MULTI-ORG, MULTI-DIE
AI-NATIVE VIRTUAL CHIPLET ECO-SYSTEMS

MULTI-ORG, MULTI-DIE
CHIPLET ECO-SYSTEMS







Specs | Planning | Design | Backend | Chiplet eco-systems

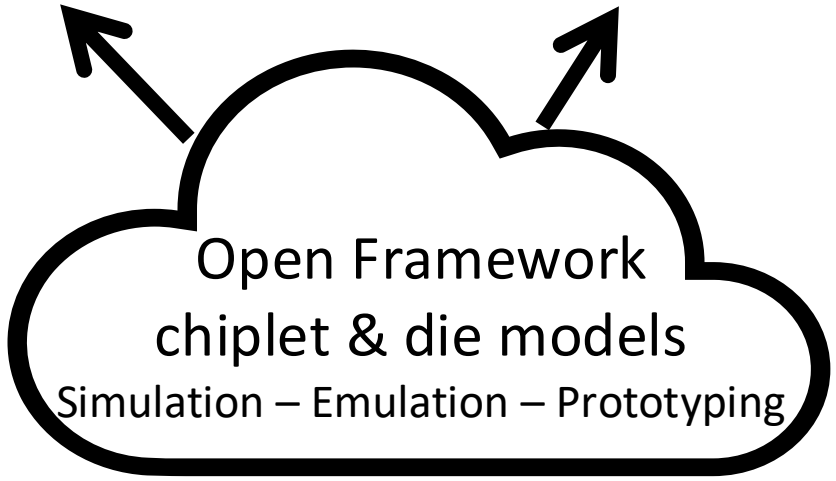
Virtual Chiplet Ecosystem = Virtual Prototyping + AI Native

Virtual Prototyping

- Distributed Simulation
- Encrypted models
- Containerized models
- Fast models
- Fixed virtual platforms
- Performance analysis
- Verification IP
- HW/SW co-design
- Reference Architectures

Design Space exploration

Tech & Business value



Standards | Governance | Security
Multi-vendor environments

AI Native

- Clouds – local and remote
- Semantic & keyword search
- Automated Interop
- Design space exploration
- Rank ordering and scoring
- Highlight gaps
- Soft IP generation
- Harness to Virt Proto tools

Panel Discussion:
AI-native Virtual Chiplet Eco-systems: Industry best practices to accelerate Chiplet adoption



Marc Meunier

Director HW Ecosystem – ARM



Lichen Weng

SOC Architect – Tenstorrent



Nandan Nayampally

CCO – Baya Systems



Matt Holdrege

Research Fellow – Microchip



Raghu Shankar

Entrepreneur – cloud and semi

Q1 Benefit and Challenge – customer

As a tech developer representing the customer perspective, what is the potential for developers investigating power, thermals, and partitioning in multi-vendor, multi-die chiplet virtual eco-systems? What is one challenge?

Matt

Q2 Benefit and challenge – SOC architect

From an SOC architect’s perspective what is the potential for chiplet virtual eco-systems to accelerate SOC design? What is one challenge?

Lichen

Q3 Potential and Challenge – Fabric

Baya Systems is uniquely positioned to bring together disparate IP through complex fabrics, applying system level modeling, simulation, and workload-driven optimization.

What do you see as potential and challenge for vendor-neutral virtual chiplet eco-systems?

Nandan

Q4 End-to-end eco-systems from IP to systems

As an ARM eco-system provider ranging from IP to systems, and list of partners and customers developing in the ARM eco-system where do you see an opportunity for pre-silicon eco-systems to validate tech readiness and accelerate time to solutions?

One challenge?

Marc

Q5 Use cases reusable chiplets/die/soft IP

In terms of use cases what’re examples of reusable chiplets/soft IP via virtual chiplet eco-systems that enables evaluation of technical and business value?

- Chiplet reuse across multiple vendors, generations or product lines?
- Out-sourcing non-strategic IP to 3rd party chiplets?
- Disaggregation of large monolithic dies to smaller chiplet/dies on heterogenous integration?
- IO die (PCIe or CPO), LPDDR, SPHBM, HBF, security chiplet, encryption chiplet? Security functions
- Heterogenous computing environments – mix of scalar and vector

Matt, Lichen, Nandan, Marc

Q6 Compliance of chiplets to Open Chiplet Atlas?

Do you see Open Chiplet Atlas evolving to simulate/emulate hardware and software components in a vendor-neutral environment?

Can partners bring in their soft IP (Behavioral or RTL models) and claim compliance or compatibility with Open Chiplet Atlas?

Lichen

Q7 Compliance of chiplets to FCSA?

Do you see FCSA evolving to simulate/emulate hardware and software components in a more open vendor environment?

Can partners bring in their soft IP (Behavioral or RTL models) and claim compliance or compatibility with FCSA?

Marc

Q8 Common elements for alignment?

Are there any common elements in these approaches that can be “unified” or made similar for easy portability of chiplet/soft IP models between virtual eco-systems?

Examples: Security, Root of Trust (RoT), and Firmware signing

Marc

Matt

Lichen

Nandan

Q9 Potential “standards” for alignment?

Are there any potential “standards” or modular verification flows in the abstraction layers, across vendors, we can start formalizing for virtual chiplet eco-system?

Choosing the right models for arch interop testing?

Matt

Q10 Pre-requisite steps for chiplet systems?

Are there any pre-requisite steps we as an industry can make today – for example methodologies or constructs – that sets the stage for virtual chiplet eco-systems?

- Examples – SystemC, Fast models, FPGA prototyping, verification IP, UVM tests, Distributed simulation, tech value, business case, ..

Marc, Lichen, Nandan

Q11 AI tools for virtual chiplet ecosystems?

One AI tool appropriate for virtual chiplet ecosystems?

Examples: Automated Interop, design space exploration, chiplet IP generation, harness to Virt Proto tools

Marc

Lichen

Nandan

Matt

Q12 Next steps to build momentum?

From process perspective what is one immediate next step to build the momentum that can be worked through OCE from Jan 2027? Who should lead and participate?

Example: Security study for alignment

Marc

Lichen

Nandan

Matt

Call to Action – virtual chiplet ecosystems

- Founder members – Build critical mass of industry leaders & contributors
- Shape the vision and outcomes in 2027
- Program Page:
  - https://www.opencompute.org/community/open-chiplet-economy/open-framework-for-chiplet-eco-system-virtual-prototyping
- Resources:
  - https://www.opencompute.org/blog/developing-ai-native-automated-virtual-chiplet-eco-systems-shift-left-shift-up-and-shift-out-to-accelerate-chiplet-adoption
  - https://www.opencompute.org/blog/accelerating-early-stage-explorations-with-virtual-prototyping-for-a-thriving-multi-vendor-chiplet-ecosystem

Open Discussion



Section: ARM



Shifting left: Faster innovation through pre-silicon integration

MARC MEUNIER, DIRECTOR HARDWARE ECOSYSTEM, CLOUD AI, ARM

ASHWIN MATTA, DIRECTOR OF PRODUCT MANAGEMENT, CLOUD AI, ARM

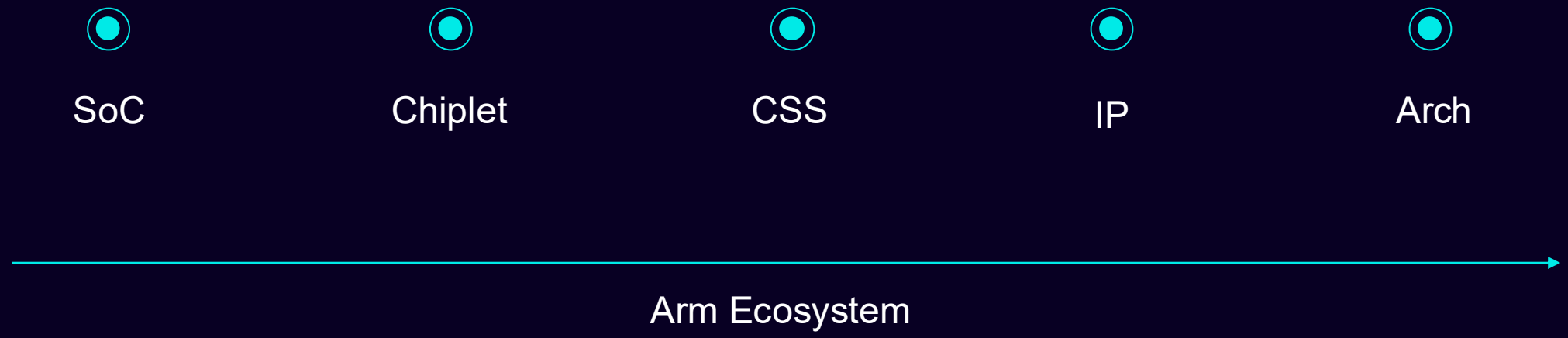
SRINI NARAYANA, VP, BOOT FIRMWARE, AMI

TRAN NGUYEN, VP, MULTI DIE ECOSYSTEM PLATFORMS SOLUTIONS, SYNOPSYS

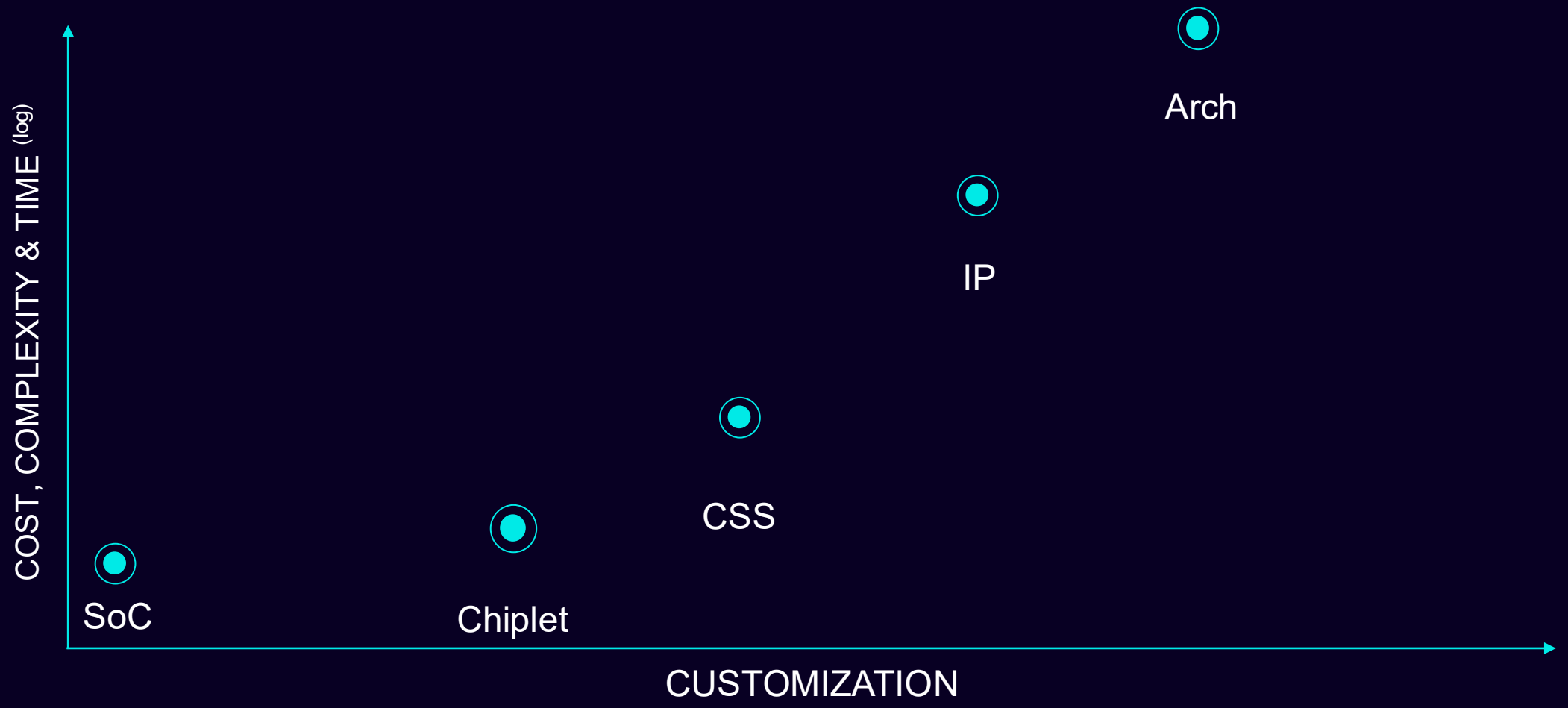
APM26

arm

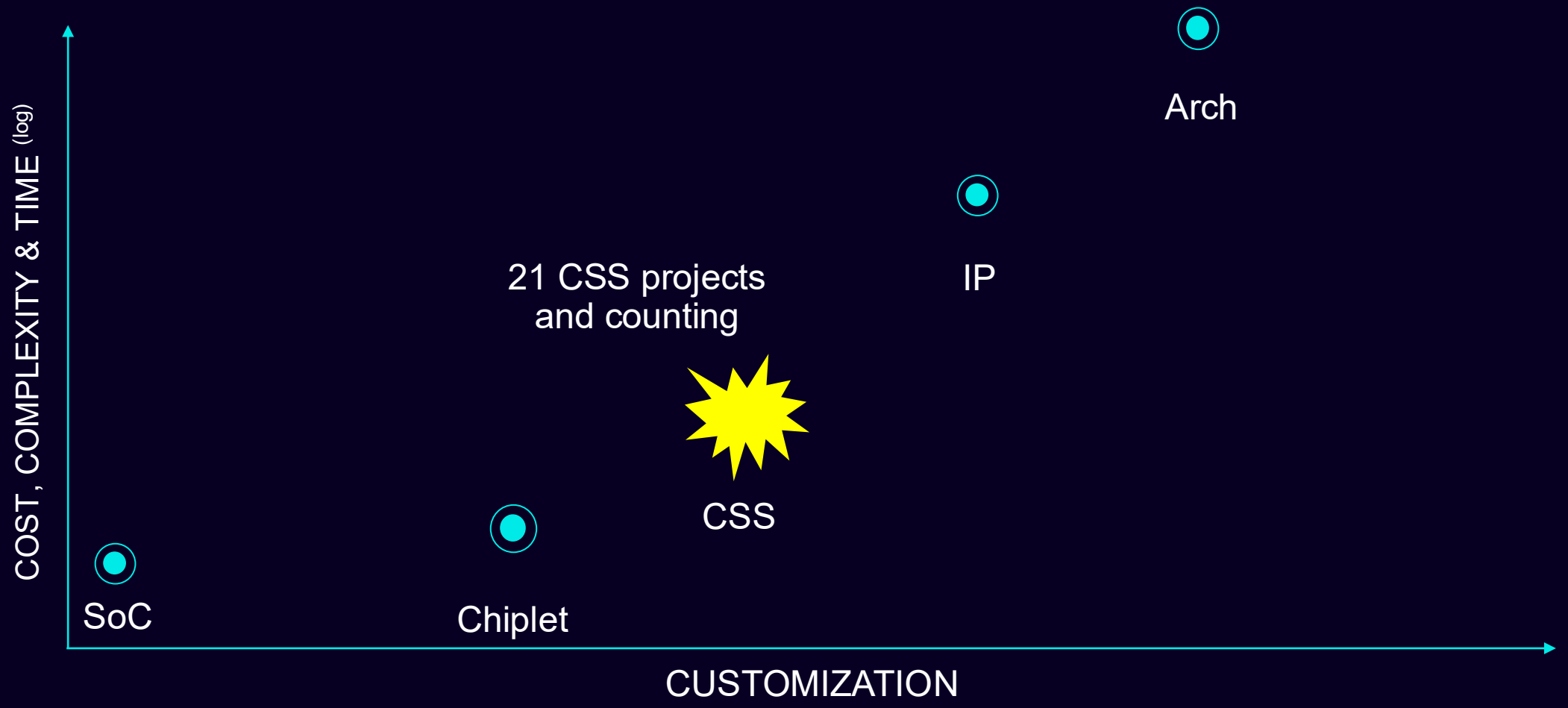
Arm offers options



Not all are created equal



Not all are created equal



Neoverse Compute SubSystem (CSS)

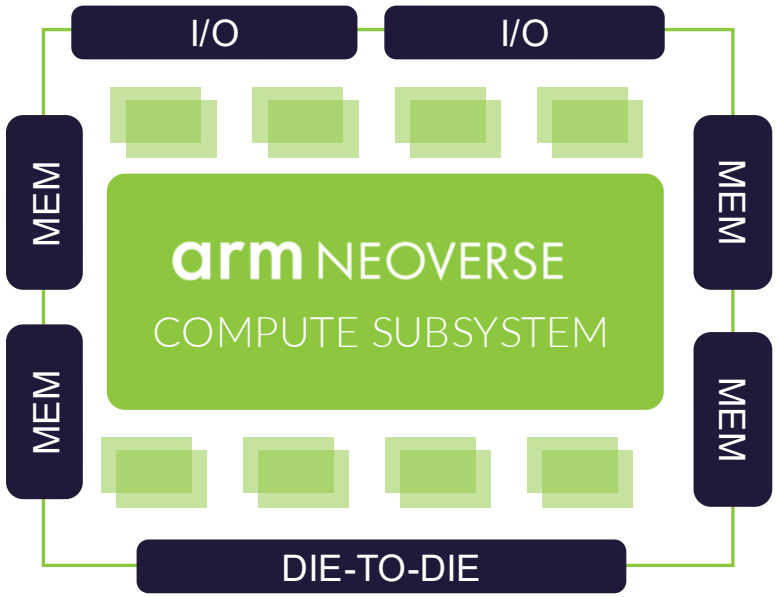
- Pre-configured Neoverse IP with licensee input for target use cases
- Includes CPU integrated with System IP and overall SoC management
- Verified, validated, and PPA-optimized by Arm
- Delivered with reference design, implementation and Day-1 ready SW stack

LATEST ARM TECHNOLOGY

CPU architecture, memory, I/O

CHIPLET READY

Arm Chiplet System Architecture (CSA)



9 months start to tape out

80 engineering years saved

arm total design

FOUNDRY SERVICES

intel foundry services

SAMSUNG

tsmc

DESIGN SERVICES & PACKAGING

ADTechnology

Alchip

ASE

ALCOR MICRO

asicland

BROADCOM

CoAsia

GUC

FARADAY

HCLTech

MARVELL

MEDIATEK

NOVATEK

REALTEK

SEMI FIVE

socionext

VIA NEXT

FIRMWARE, EDA & 3RD PARTY IP

ami

cadence

CEVA

insyde

proteanTecs

Rambus

SYNOPSYS

PUFsecurity

SIEMENS

COMPANION SILICON & OEMs

AsteraLabs

CREDO we connect.

eliyan

Jaguar

NOKIA

rebellions_

SIPEARL

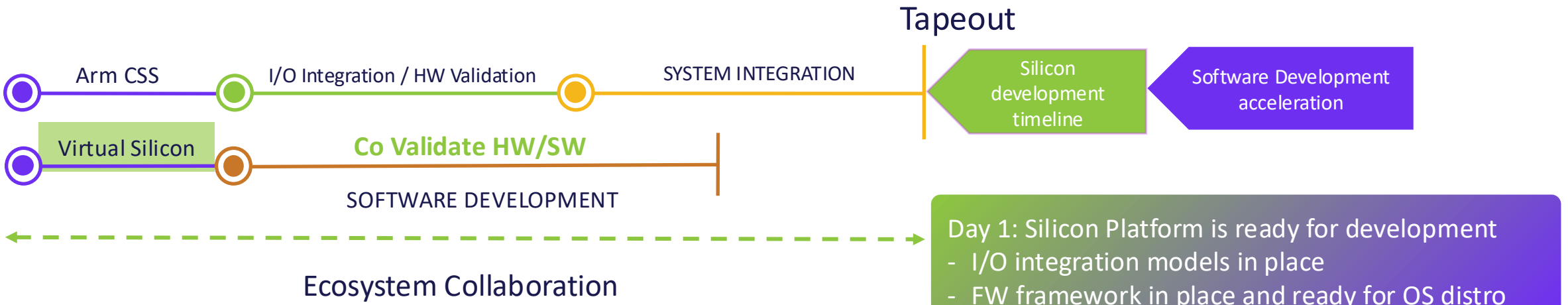
CONFIDENTIAL © 2026 ARM

ARM Virtual Silicon: Faster time to Market. Reduce Risks

Original development process



New*: Reduced Time-to-silicon and Accelerated Software Development



Day 1: Silicon Platform is ready for development
- I/O integration models in place
- FW framework in place and ready for OS distro

ARM: How are we going to engage ecosystem partners?

01

Provide early access to next-gen CSS

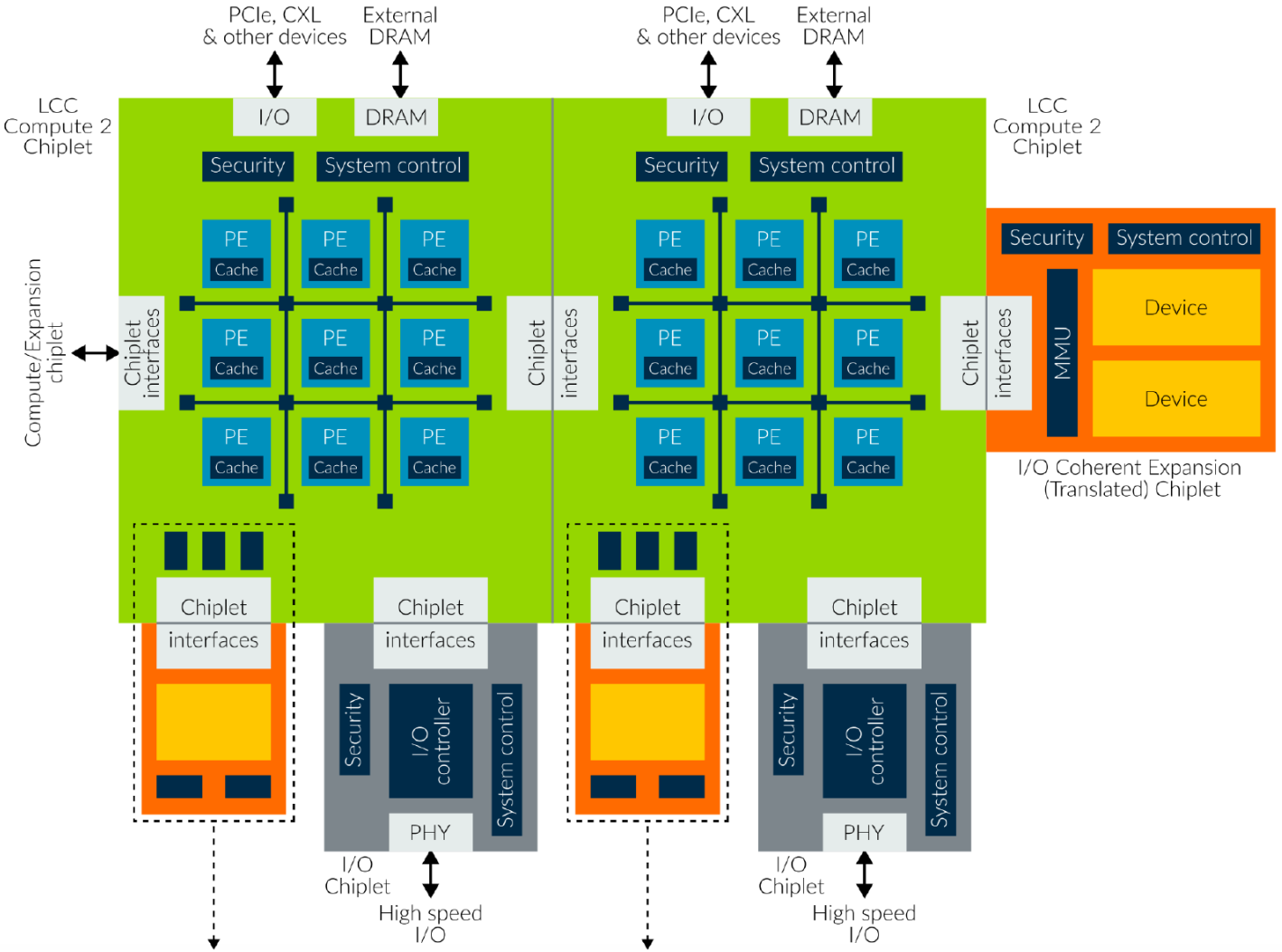
02

Integrate 3rd party IP

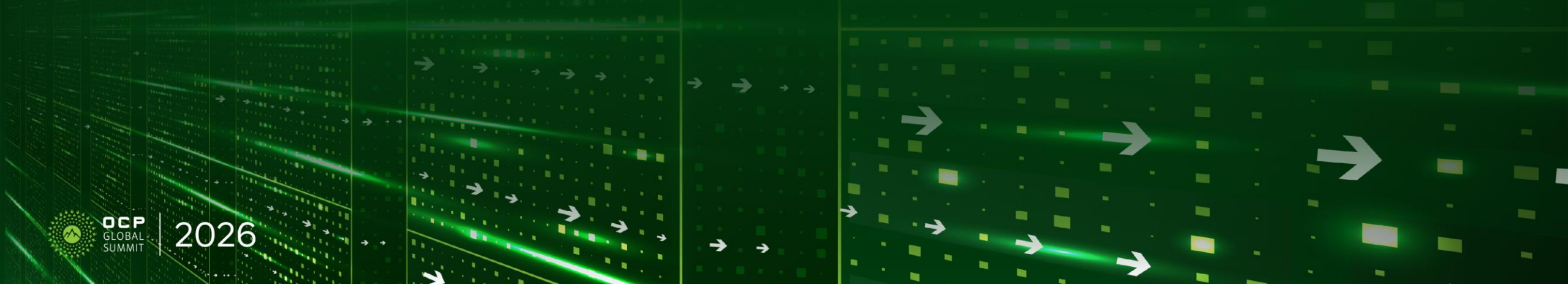
03

Publish guides and validation results

ARM – Enabling Chiplet Reuse and Composable SoC's through Standardization; *source Mark Knight, Chiplet Summit*



Section: Tenstorrent



Challenges

- Tenstorrent and partners build high-performance semiconductors using independently designed chiplets
- Need for an **immediate solution** to multi-vendor chiplet integration
- Our engineering practice revealed the following main challenges

Lack of a full-stack chiplet interoperability standard

- Lack of open definition to map bus protocols to D2D
- Incomplete definition to make system-level features interoperable
- In need of standard for packaging

Lack of compliance tests

- High development risk

Lack of reusable chiplets

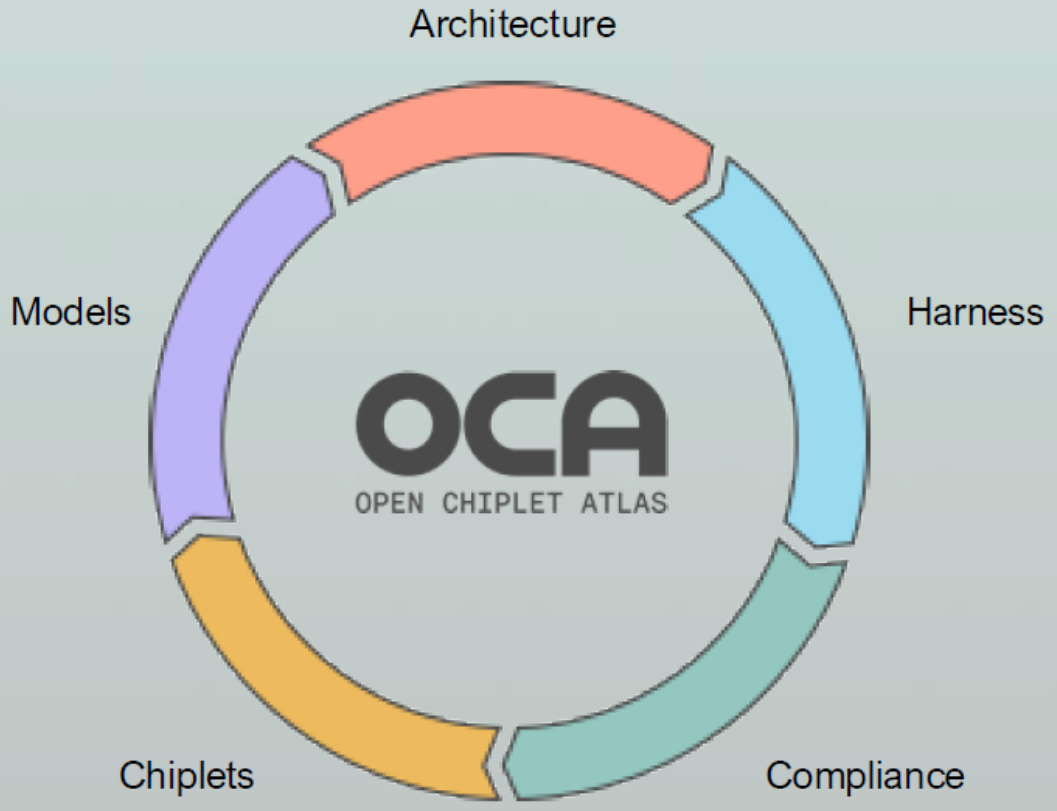
- Discourages adoption of the chiplet based method, which further contributes to scarcity of reusable chiplets

Lack of open-source reference design

- Spec ambiguity at the early stage of standard definition
- Need for an open-source reference design to complement the spec

Open Chiplet Atlas® Ecosystem

An ecosystem that favors speed, productization, and immediate availability



OCA Architecture. Open architecture for chiplet interoperability, no vendor lock-in, ISA-neutral.

OCA Harness. Open-source design of a chiplet framework, a collection of all non-application logic.

OCA Compliance. OCA Architecture compliance test suite for both pre- and post-silicon.

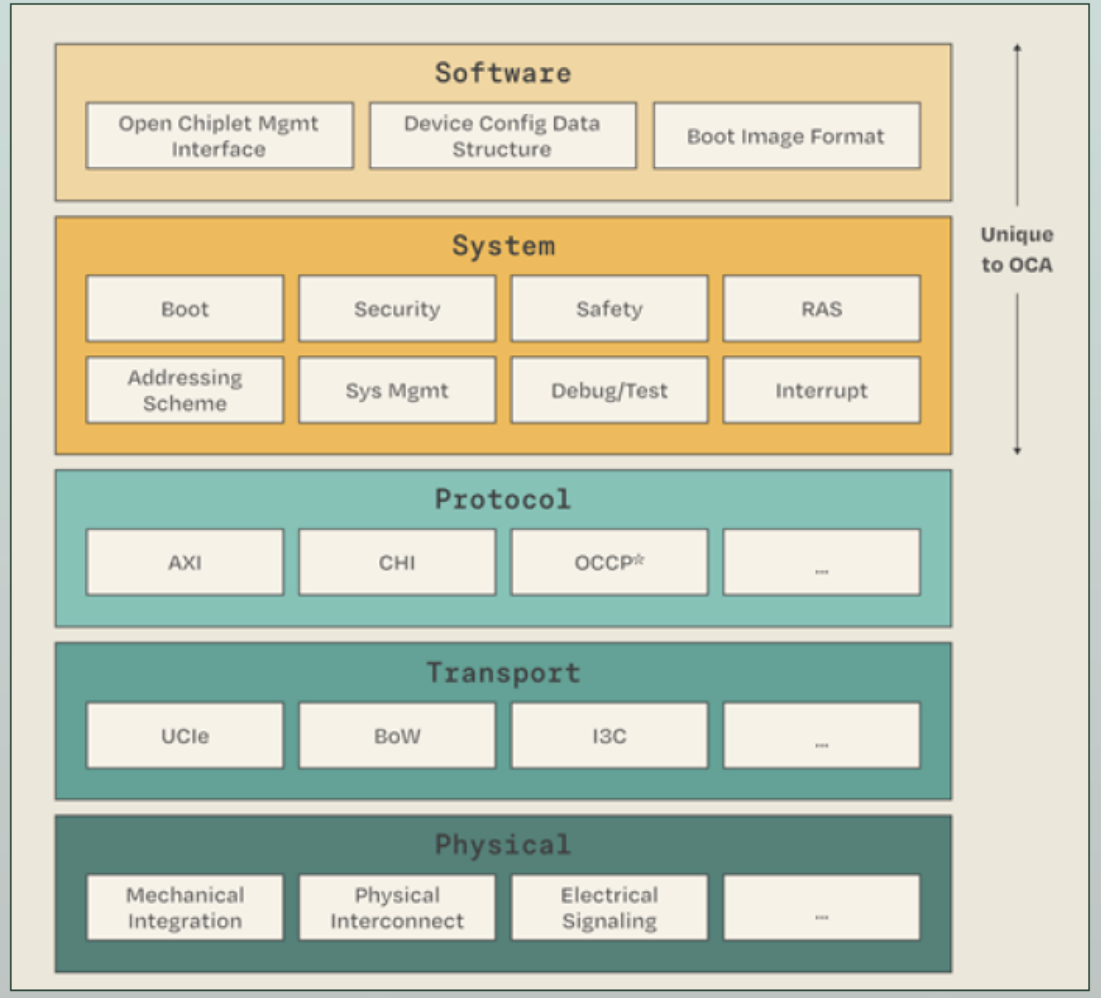
OCA Chiplets. OCA backed up by Tenstorrent chiplets (AI, CPU, memory, IO).

OCA Models. Function and performance models for the OCA Harness.

OCA Architecture

An architecture grounded in proven, production silicon

- A complete 5-layer solution to chiplet interoperability
- Vendor lock-in free, ISA neutral
- Product-based, silicon proven
- Permissive license for both adoption and modification
- Maintained by Tenstorrent, open to feedbacks



Available now on openchipletatlas.org:

Draft v0.8 of the OCA System Arch Specification

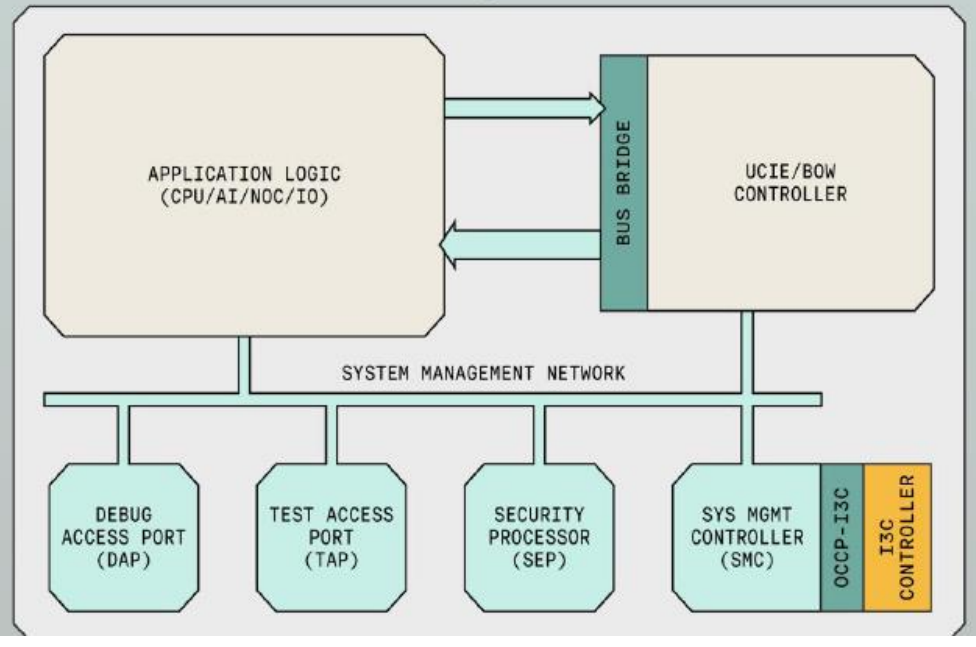
Draft v0.8 of the OCA AXI over UCIe (AoU) Specification

OCA Harness: System IPs and D2D Solutions

Open source chiplet framework as the OCA reference design. Seamless integration into partners' designs

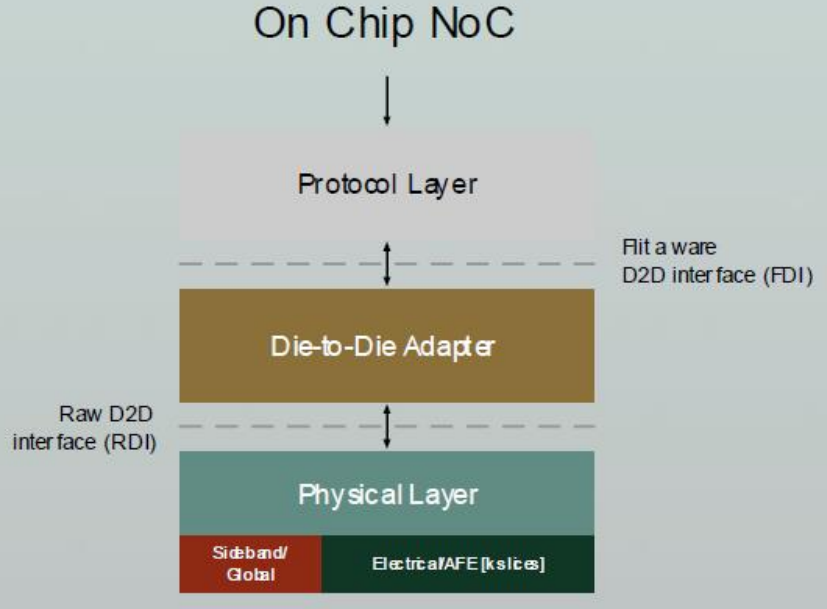
Harness Hardware components

- Security Processor
- System Management Controller
- Design and Test Ports (DTP)
- System Management Network
- AXI over UCIe Bridge



Harness Software components

- Boot Loader - BL0 (ROM), BL1, BL2
- SEP/SMC runtime RTOS
- SEP/SMC services (e.g. attestation)

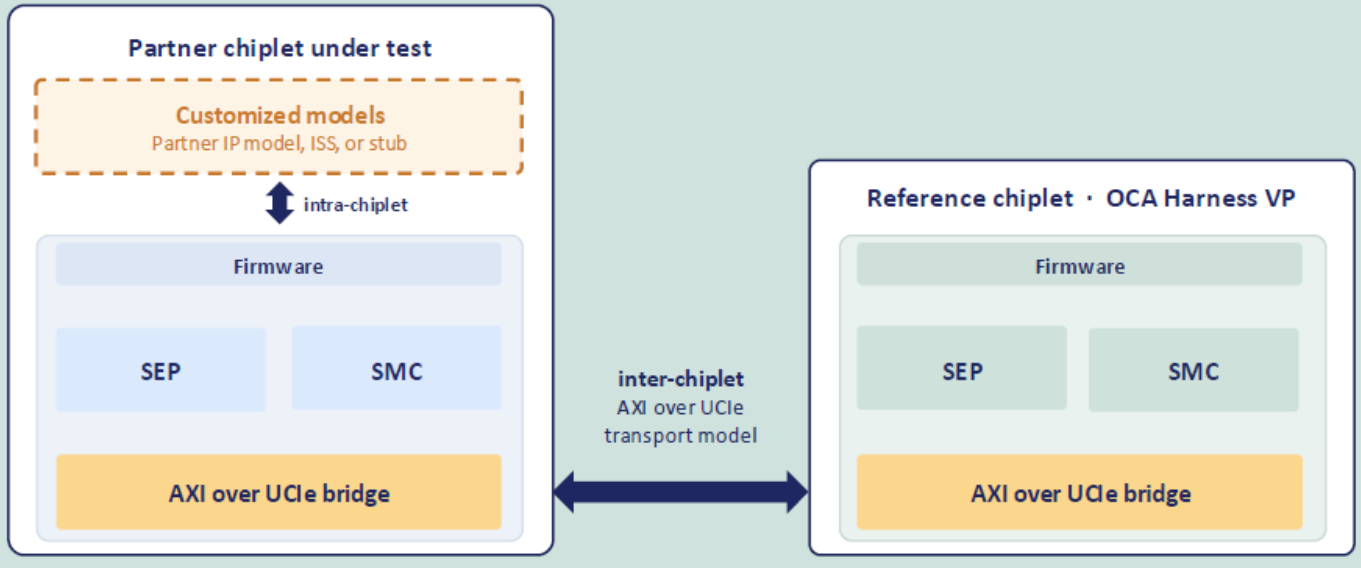


Beta Release Oct 2026 under Apache 2.0

OCA Models: Virtual Platform

Bring Your Own Models

Functional model · loosely timed



Cross-chiplet boot, I3C BL1 recovery	Multi-chiplet mgmt and communication	OCTS time sync across chiplets	Inter-chiplet transactions over AXI routing over UCIe

Behavioral compliance only · timing, PHY and packaging live in their own specs

From Seed to Forest

3 Phase Execution

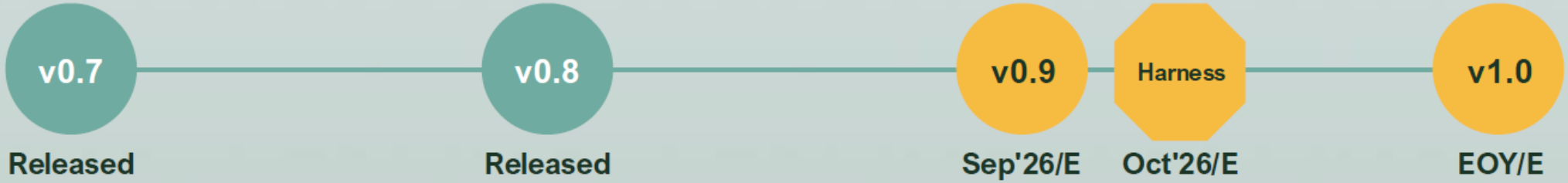
- **Phase 1 · IP / unit** — SEP + SMC IPs, Zephyr boots on the cluster **Oct'26**
- **Phase 2 · Chiplet** — SEP + SMC + AXI over UCIe in one Harness **Oct'26**
- **Phase 3 · Multi-chiplet** — 2+ Harness instances over the transport by EOY/E

Where Partners Come In

- **Co-sim your logic early** — catch integration bugs before your RTL freezes
- **Report model-vs-spec gaps** — contribute to improve the ecosystem
- **Contribute test content** — cases that exercise the Harness contract, feeding OCA Compliance

OCA Roadmap

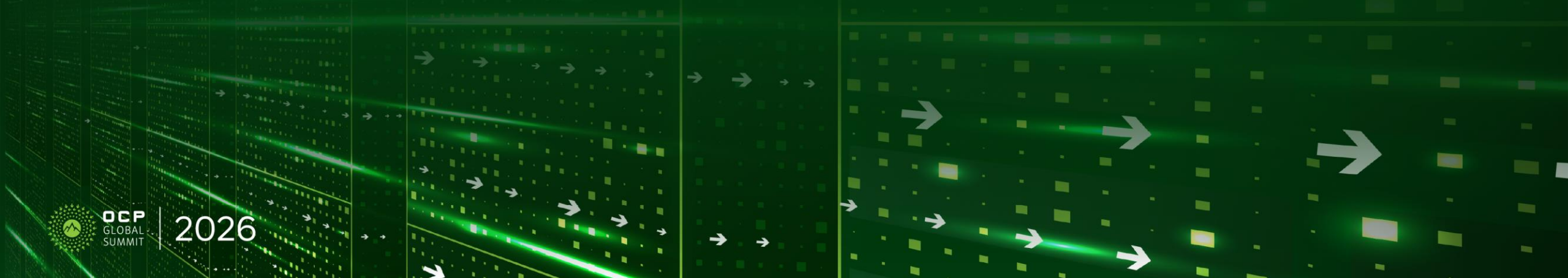
Goal: A silicon proven specification at v1.0



Topics for v0.9 and future revisions

- Open chiplet management interface: the software/firmware API for multi-chiplet management
- Post-Si feedback from TT internal chiplets and partners' implementations
- External protocol scrub: review several leveraged protocols like UCIe and CHI-C2C for the 2nd level details
- Partner feedback from ongoing collaborations and contributions

Section: Baya Systems



Key challenges in the path to scale

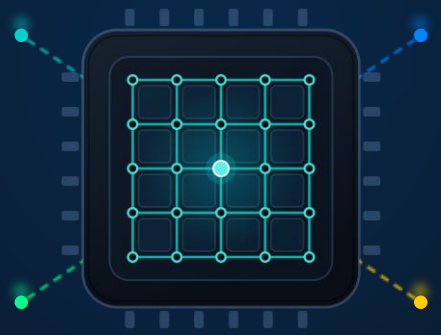
Data movement is now the bottleneck

Chiplets & multi-die

Partitioning, D2D links and cross-die coherency add system-level risk

Shoreline scarcity

SerDes, UCIe and HBM PHYs compete for a finite die edge as bandwidth soars



AI scale-up & scale-out

Scaling Crossbars for high-radix UALink, UEC, ESUN switches is not chiplet friendly

Late closure & respin risk

PPA, timing and multi-die verification issues found late drive schedule slips and respins

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End to End: Specification to Closure

One software-driven flow — from first spec to successful product in first silicon

End to End System Design — WeaverPro

01 Requirements Capture
Fast Programmatic Design

02 Workload Analysis
Faster Performance Closure

03 Fabric Design
Comprehensive, Scalable, Correct

04 Silicon Deployment
Post-silicon Tuning

CacheStudio

FabricStudio

SystemC API

Python API

Comprehensive Fabric Portfolio — WeaveIP

Cache-Coherent Fabric
Coherency CHI fabric that scales — multi-level, nested clusters, across chiplets

Non-coherent Fabric
Extremely flexible AXI NoC with a highly efficient implementation

Custom AI Fabric
Software-defined: carry any protocol at extreme AI bandwidth

Scale-Up/Scale-Out Fabric (NeuraScale)
Non-blocking crossbar replacement — simplifies AI scaling & physical design

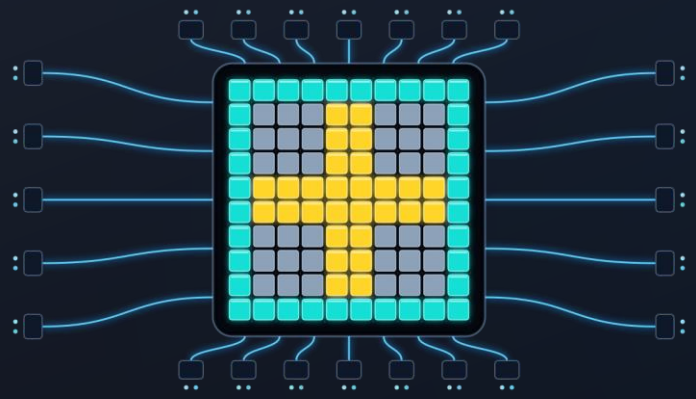
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Scale Is Won at the Shoreline

Reticle limits add serious challenges to scaling high-radix switches

Digital core inside, SerDes & UCIe on the edge



Digital core | SerDes /PHY/IO | UCIe D2D

What sets NeuraScale apart

Crossbar	Mesh NoC	NeuraScale
✓ Non-blocking performance	✗ Non-blocking performance	✓ Non-blocking performance
✗ Scales to 1000s of ports	✓ Scales to 1000s of ports	✓ Scales to 1000s of ports
✗ Frees edge for IO/SerDes	✓ Frees edge for IO/SerDes	✓ Frees edge for IO/SerDes

1 PHY
multi-protocol D2D link

320 GB/s
per UCIe D2D link at 2.5 GHz

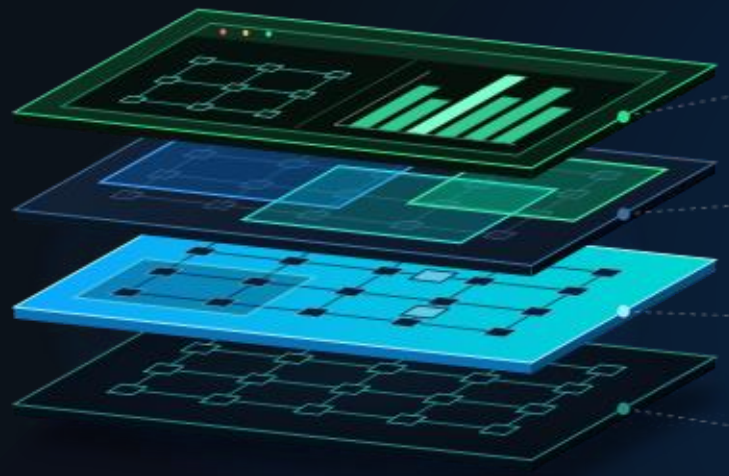
256 1Tb/s ports
per chiplet · fully non-blocking

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Shift Left, Built In

Flexible fabrics, early analysis and verification — decided at spec time. Full system view



ANALYSIS

Analyze Before RTL

Workload & performance closure at spec time

FLEXIBLE

Any Topology · Any Protocol

PREDICTABLE

Generated, Human-Readable RTL

Change the spec, regenerate the fabric — no hand edits

QoS | Multicast | RAS | FuSa | Width conversion | Physical design and tiling

CONNECTED

Chiplet-Ready

D2D streaming packetizer, PHY-aware

PHYSICAL

Floorplan-Aware

Placement & DEF guidance, day one

VERIFIED

Correct Across Dies

Cross-die simulation & models, deadlock free design

Analyze at spec time
closure before RTL

Iterate in software
Python & SystemC APIs

Fewer late surprises
deadlock-free across D2D, lower respin risk

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What next-generation scale looks like

32+ chiplets in package

16×16 mesh per die

2048-bit links



Streaming connectivity through Baya D2D packetizer

100s of agents

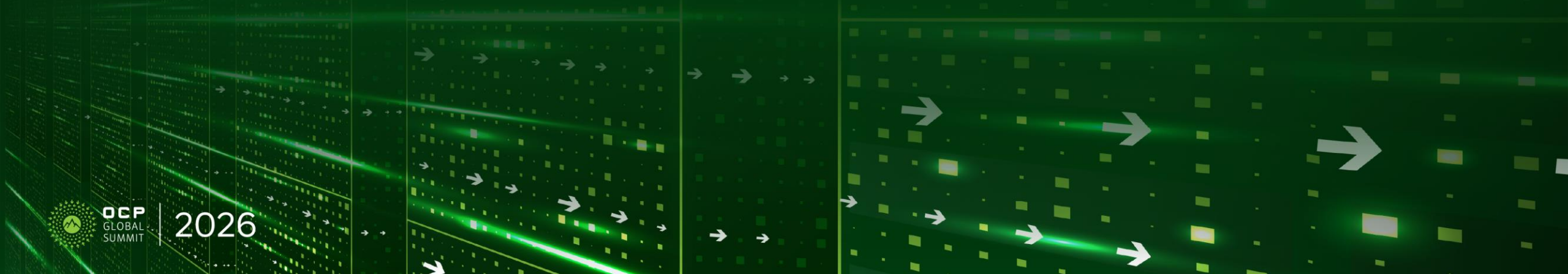
Many parallel wires per router-to-router link

>10× larger scale solutions	256 TB/s aggregate throughput / chiplet	30:1 reduction in scale-out switches

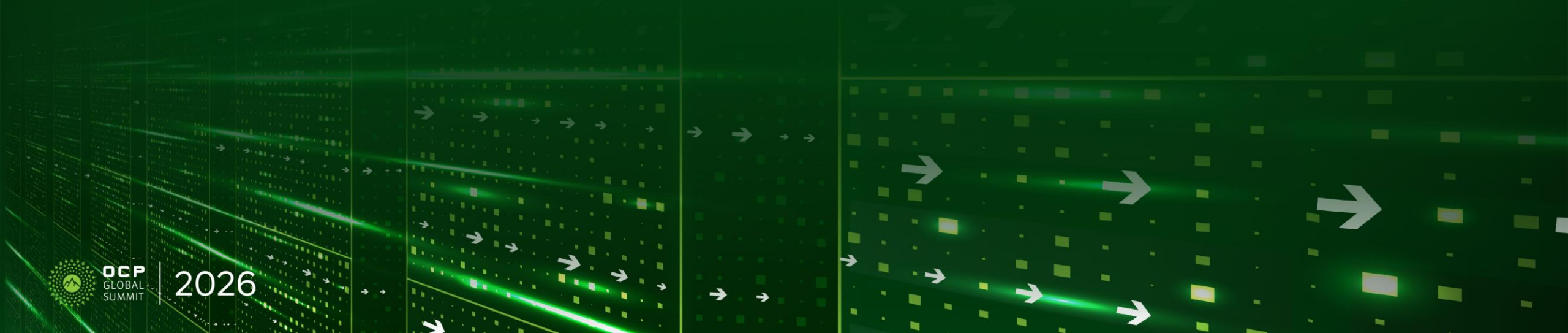
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Section: Microchip



Abstract & Panelists



Abstract – Virtual chiplet eco-systems

- Despite the advantages of chiplets, broad adoption addressing a wider range of market opportunities by smaller organizations remains a major challenge.
- We need to reduce friction at the front-end of design cycle that includes architecture and business planning thereby increasing design space explorations in multi-vendor environments.
- Define AI-native virtual ecosystems built on industry best practices of virtual prototyping that will accelerate chiplet project evaluations.
- AI must be applied extensively to increase early-stage explorations while reducing mundane and repetitive tasks. These include semantic and keyword search of specs, rank ordering and scoring top matches, highlighting gaps, code generation, harness to EDA tools, test benches, and benchmarks.
- Incorporate best practices including encrypted models, containerized models, fast models, fixed virtual platforms, cloud, local and remote verification IP for interoperability validation, IP libraries, and scenario-based modeling.

Marc Meunier



- Director Hardware Ecosystem – ARM
- Leads ecosystem and industry collaboration efforts focused on custom silicon, chiplet-based architectures, and accelerated platform development through Arm Total Design.
- His work spans next-generation server and datacenter technologies, helping partners bring innovative Arm-based solutions to market faster.
- Marc is particularly focused on enabling the infrastructure required for AI-driven and agentic computing workloads.

Lichen Weng



- Lichen Weng is an SoC Architect at Tenstorrent, working on next-gen chiplet-based products and the Open Chiplet Atlas® (OCA) architecture.
- Prior to Tenstorrent, Lichen spent several years as an SoC Architect at Intel, where he contributed to multiple generations of data center CPUs.
- Lichen holds a Ph.D. in Computer Engineering from Florida International University, with research specializing in dynamic scheduling and resource management for multicore, multithreaded systems.

Nandan Nayampally



- Chief Commercial Officer of Baya Systems. He is a semiconductor executive with nearly three decades of experience in engineering, product and business roles.
- Extensive background in processing, systems and software and in particular IP, after an initial start on Athlon processor team at AMD, he spent 16 years at Arm building the processor, fabric, vertical product businesses, and was General Manager for the CPU business and first GM of the Client business.
- He led Amazon's 3rd party Alexa effort, and had key roles at Denali, and most recently at BrainChip prior to his role at Baya Systems, where he focused on bringing disruptive technologies to market.

Matt Holdrege



Fellow at Microchip Technology with

25+ Years in Strategic Technology Development and Implementation

Raghu Shankar



- Entrepreneur: cloud native and chiplets
- Dell: Server product management – HPC & AI
- Incubation and business development of AI
- Intel: Architect and Product Management
- https://www.linkedin.com/in/raghu-shankar/